

# LKA Invoices

Environment setup: live (production) and test (staging), step by step for every tool

Lingeshwar Kaparthy & Associates · Code: [github.com/adi-repo-ai/ca-invoice-app](https://github.com/adi-repo-ai/ca-invoice-app) · Updated October 2026

|                  | Live (production)                             | Test (staging)                                                                                                                     |
|------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| GitHub branch    | <code>main</code>                             | <code>staging</code>                                                                                                               |
| Netlify address  | <code>lingeshwarca-invoice.netlify.app</code> | <code>staging--lingeshwarca-invoice.netlify.app</code> · previews: <code>deploy-preview-N--lingeshwarca-invoice.netlify.app</code> |
| Firebase project | <code>ca-invoice-app</code> (real data)       | <code>ca-invoice-app-staging</code> (made-up data only)                                                                            |
| Netlify credits  | 15 per deploy (free plan: 300 / month)        | 0 (branch deploys and previews are free)                                                                                           |
| How to tell      | no banner                                     | yellow "TEST SITE — not real data" banner                                                                                          |

## 1. GitHub (code)

1. **Branches:** `repo` → branch dropdown (shows `main`) → type `staging` → **Create branch: staging from main** (already done).
2. **Release** (= "deploy"): **Pull requests** → **New** → base `main` ← compare `staging` → Create → **Merge**. To update `main` without a live build, put `[skip netlify]` in the merge commit title.
3. After a release, bring `staging` level with `main` (pull request `main` → `staging`, or "Update branch").

## 2. Firebase (data and sign-in) — create the test project once

1. `console.firebase.google.com` → **Add project** → `ca-invoice-app-staging`, Analytics off, **Spark** (free) plan.
2. **Build** → **Authentication** → **Get started** → enable **Google** and **Email/Password**.
3. **Build** → **Firestore Database** → **Create database** → production mode, same region as live. (If it says "already exists", it is done; never create a second named database.)
4. **Authentication** → **Settings** → **Authorized domains** → **Add** `staging--lingeshwarca-invoice.netlify.app` (two dashes).
5. ⚙️ **Project settings** → **General** → **Your apps** → **Web (</>)** → register "staging" → copy `apiKey`, `authDomain`, `projectId`, `appId`.
6. ⚙️ **Project settings** → **Service accounts** → **Generate new private key** → a `.json` file downloads. **Secret:** never email, chat or commit it.
7. Recommended for both projects: ⚙️ **Users and permissions** → add a second Owner (account-loss protection).

## 3. Netlify (website and server functions)

1. **Site configuration** → **Build & deploy** → **Branches and deploy contexts** → **Configure:** Production branch `main`; Branch deploys → *Let me add individual branches* → `staging`; Deploy Previews → *Any pull request against your production branch* (optional).
2. **Site configuration** → **Environment variables** → each variable → **Edit** → **Different value for each deploy context:**

| Variable                                                                                                      | Production                  | Branch deploys + Deploy Previews                                                  |
|---------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------|
| <code>VITE_FIREBASE_API_KEY</code> / <code>AUTH_DOMAIN</code> / <code>PROJECT_ID</code> / <code>APP_ID</code> | live web config             | test web config (step 2.5)                                                        |
| <code>FIREBASE_PROJECT_ID</code>                                                                              | <code>ca-invoice-app</code> | test project id                                                                   |
| <code>FIREBASE_CLIENT_EMAIL</code> / <code>FIREBASE_PRIVATE_KEY</code> (secret)                               | live service-account key    | <code>client_email</code> / <code>private_key</code> from the test key file (2.6) |

|                               |       |                                                               |
|-------------------------------|-------|---------------------------------------------------------------|
| SETUP_CODE (secret, optional) | empty | any text of 12+ characters, only until the first owner exists |
|-------------------------------|-------|---------------------------------------------------------------|

3. **Deploys** → **Trigger deploy** on the `staging` branch (or push to it) so the new values are used. Never add `VITE_USE_FIREBASE_EMULATORS` or `..._EMULATOR_HOST` here.

## 4. Cloud Shell (database rules and indexes)

shell.cloud.google.com, signed in with the Google account that owns the projects:

| When                                             | Commands                                                                                                                                                                                                           |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First time                                       | <code>git clone https://github.com/adi-repo-ai/ca-invoice-app.git</code><br><code>&amp;&amp; cd ca-invoice-app &amp;&amp; npm install</code>                                                                       |
| Test project (after rule changes on staging)     | <code>cd ~/ca-invoice-app &amp;&amp; git checkout staging &amp;&amp; git pull</code><br><code>npx firebase deploy --only firestore:rules,firestore:indexes</code><br><code>--project ca-invoice-app-staging</code> |
| Live project (after a release with rule changes) | <code>cd ~/ca-invoice-app &amp;&amp; git checkout main &amp;&amp; git pull</code><br><code>npx firebase deploy --only firestore:rules,firestore:indexes</code><br><code>--project ca-invoice-app</code>            |

Success ends with ✓ **Deploy complete!**. New indexes take a few minutes to build; the app shows "still preparing this view" meanwhile.

## 5. First owner and sample data on the test site

1. **Owner, option A:** open the test site → Sign in with Google → "No access yet" page → enter the test `SETUP_CODE`. Then delete `SETUP_CODE` from Netlify.
2. **Owner, option B (no code):** upload the test key to Cloud Shell, then `npm run create-admin -- --key ~/staging-key.json --email you@gmail.com --password 'TempPass123' --name 'Your Name' and rm ~/staging-key.json.`
3. Settings → fill in and **save firm details**.
4. Settings → **Test site: sample data** → **Load sample data** (6 TEST clients, 10 invoices). Only owners see it, only on test addresses.

## 6. Release checklist (every change)

1. Developer pushes to `staging` (tests + build pass first).
2. If rules changed: deploy them to the **test** project (4).
3. Owner tests on the test site as Owner / Admin B / Staff.
4. Owner says "deploy" → pull request `staging` → `main`.
5. Optional: check the Deploy Preview link on the pull request.
6. **Merge** → Netlify publishes live in ~2 minutes.
7. If rules changed: deploy them to the **live** project (4).
8. Problem? Netlify → Deploys → previous deploy → **Publish deploy** (instant rollback).

**Safety:** real client data only on live; keys and setup codes never in chat, email or GitHub; keep 2-step verification on Google, GitHub and Netlify; Firebase stays on the free Spark plan for both projects.